

the start-up “to launch our vehicles from a facility we have designed and built”. Agrees Co-founder Moin S.P.M. “The reforms truly accommodate everyone’s dream of going to space.” If all goes well, AgniKul will soon become the second start-up to put its launch vehicle in space.

“Skyroot and AgniKul together make for a very good offering. It will probably be a year before they launch, from which point, they [the start-ups] will take off. For the Indian space sector, that will be a great achievement as there will be two private companies competing globally,” says Goenka. He adds that the addition of ISRO’s own Small Satellite Launch Vehicle (SSLV) creates a very robust small satellite launch infrastructure in India.

While upstream in the space sector is about sending objects such as rockets and satellites into space, downstream utilises the research and data generated by upstream in a range of applications. Industry experts like Lt Gen. (Retd) Anil Kumar Bhatt, Director General of Indian Space Association (ISpA)—the apex body of private players in the sector—point out that a significant portion of growth opportunities lie in the downstream sector. “The largest percentage of opportunities in the sector will be in applications. There is no line to draw there since that includes everything from communications, health, education, and mining to intelligence, surveillance and reconnaissance (ISR) for the armed forces... Although rockets and satellites are the core, they only form a minuscule percentage of the space economy,” he says.

Agrees Gunjan Thakuria, Founder & CEO of Navi Mumbai-based start-up Tathya Earth, which offers predictive economic forecasts based on satellite imagery. “As the cost of data decreases and the types of data increase, various ap-

plications critical to diverse industries and governments will evolve,” he says. That also means that the sector will need more people with skills to turn upstream research into downstream applications.

With a 36 per cent share, the satellite services and applications segment will form the largest share of India’s space economy in 2025, says a joint study released by EY and ISpA in October. The report projects the country’s space economy expanding to \$13 billion in 2025, with the space launch segment growing at a CAGR of 13 per cent by that year, which will be further spurred through enhanced private participation, the latest tech and low cost of launch services. Moreover, the country’s satellite manufacturing capability is expected to be worth \$3.2 billion by 2025 from \$2.1 billion in 2020. “Satellite manufacturing, both for use within and outside India, will become fairly large. And IN-SPACe is in discussions with two state governments to create a manufacturing hub to incentivise the setting up of satellite production facilities as well as providing technical assistance,” says Goenka.

Satyanarayanan Chakravarthy, Professor of Aerospace Engineering at IIT Madras and Co-founder of transport tech firm The ePlane Company, sees India emerging as the carrier of the world’s payloads into outer space. “Along with SSLV of ISRO, Skyroot and AgniKul put together, we can potentially become the space lorries, if you will, to haul all the baggage for the world,” he says. “The next thing that can happen is that start-ups can piggyback on ISRO’s success and double down in taking humans to at least LEO for tourism. The potential is huge.”

INVESTORS CLAMOUR

Spacotech, however, is a capital-intensive business. The good news is



“The space sector had to be opened up because we needed to encourage people in the private space to contribute to the domains where there was a business opportunity”

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that the quantum of funding raised by such start-ups is on the rise. Besides, with an increasing number of proofs of concept ready, it is expected to get a further boost. “Private space as a concept is new to India. So, that has taken some amount of time for investors to understand how it will work, where the market opportunities are and how will companies start making revenues,” says Vinod Kumar Shankar, Co-founder & Partner of Java Capital, a pre-seed/seed fund. “The last four years were largely spent looking for small early-stage funding, with larger funding continuing to be a challenge in the deep-tech sector. Fortunately, the government opening its arms to private sector participation through IN-SPACe has made it much more attractive.”

Goenka adds that potential in-